

Quick Activity 1: School Supplies Calculator

A notebook costs ₱45, and a pen cost ₱12. A student buys several notebooks and pens and pays a certain amount. Develop an algorithm that calculates the **total cost** and **change**.

Assume that quantities are positive whole numbers and the payment is enough to cover the purchase

Input	Process	Output
Number of notebook, number of pen, payment	Calculate notebook cost, calculate pen cost, add for total cost, calculate change	Total cost and change

Descomposition (Task)

1. Read the number of notebook.
2. Read the number of pen.
3. Read the payment amount.
4. Calculate the notebook cost.
5. Calculate the pen cost.
6. Calculate the total cost.
7. Calculate the change.
8. Display the total cost and change.

Pseudocode

START

```
READ notebook, pen, payment
notebookCost ← notebook × 45
penCost ← pen × 12
totalCost ← notebookCost + penCost
change ← payment - totalCost
DISPLAY totalCost DISPLAY change
```

END

Dry Run Calculations

Test Case 1 Input:

- Notebook: 3
- Pen = 2
- Payments = ₱200

Step	Calculation	Solution
Notebook Cost	3×45	₱ 135
Pen Cost	2×12	₱ 24
Total Cost	$135 + 24$	₱ 159
Change	$200 - 159$	₱ 41

Output

Total Cost = ₱159

Change = ₱41

Test Case 2

Input:

- Notebook = 2
- Pen = 3
- Payment = ₱150

Step	Calculation	Solution
Notebook Cost	2×45	₱ 90
Pen Cost	3×12	₱ 36
Total Cost	$90 + 36$	₱ 126
Change	$150 - 126$	₱ 24

Output

Total Cost = ₱159

Change = ₱41

Final Answer

Test Case	Total Cost	Change
1	₱159	₱41
2	₱126	₱24

